18.100B: Real Analysis

Year: Spring 2025

Instructor(s): Prof. Tobias Holck Colding

OCW URL: https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/

Textbooks

[Main] Thomson, Bruckner & Bruckner, Elementary Real Analysis, 2nd ed, 2008

[Supplementary] Rudin, Principles of Mathematical Analysis, 3rd ed, 1976

Available Resources

- 23 Lecture Notes (PDF)
- 23 Video Lectures
- 10 Problem Sets
- 1 Midterm Exam + 1 Final Exam

#	Topic	Readings	Resources
PART I: Real Numbers & Sequences			
Lec 1	Introduction to Real Numbers	Sec. 1.1-1.4	Notes PDF / Video / PS1
Lec 2	Introduction to Real Numbers (cont.)	Sec. 1.5-1.6	Notes PDF / Video
Lec 3	How to Write a Proof; Archimedean Property	Sec. 1.7	Notes PDF / Video
Lec 4	Sequences; Convergence	Sec. 2.1-2.2, 2.4-2.7	Notes PDF / Video
Lec 5	Monotone Convergence Theorem	Sec. 2.8, 2.9, 2.11	Notes PDF / Video / PS2
Lec 6	Cauchy Convergence Theorem	Sec. 2.12	Notes PDF / Video
Lec 7	Bolzano-Weierstrass Theorem; Cauchy	Sec. 2.11-2.12, 3.4	Notes PDF / Video / PS3
Lec 8	Convergence Tests for Series; Power Series	Sec. 3.5-3.6	Notes PDF / Video
Lec 9	Limsup and Liminf; Power Series; Continuous	Sec. 2.13, 10.2, 5.4, 5.5	Notes PDF / Video / PS4
Lec 10	Continuous Functions; Exponential Function	Sec. 5.5, 5.9	Notes PDF / Video
PART II: Metric Spaces & Differentiation			
Lec 11	Extreme and Intermediate Value Theorem; Metric	Sec. 5.4, 13.1-13.2	Notes PDF / Video / PS5
--	Midterm Review	--	Review Notes PDF
--	Midterm Exam	--	Exam (in-class, 1.5 hr)
Lec 12	Convergence in Metric Spaces; Operations on	Sec. 13.4-13.5	Notes PDF / Video
Lec 13	Open and Closed Sets; Coverings; Compactness	Sec. 13.5	Notes PDF / Video / PS6
Lec 14	Sequential Compactness; Bolzano-Weierstrass in	--	Notes PDF / Video
Lec 15	Derivatives; Laws for Differentiation	Sec. 7.2, 7.3	Notes PDF / Video
Lec 16	Rolle's Theorem; Mean Theorem; L'Hopital's Rule;	Sec. 7.6, 7.11-7.12	Notes PDF / Video / PS7
PART III: Integration & Convergence			
Lec 17	Taylor Polynomials; Remainder Term; Riemann	Sec. 8.6	Notes PDF / Video
Lec 18	Integrable Functions	Sec. 8.3	Notes PDF / Video / PS8
Lec 19	Fundamental Theorem of Calculus	Sec. 8.3	Notes PDF / Video
Lec 20	Pointwise Convergence; Uniform Convergence	Sec. 9.2-9.4	Notes PDF / Video / PS9
Lec 21	Integrals and Derivatives under Uniform	Sec. 9.5-9.6	Notes PDF / Video
Lec 22	Differentiating and Integrating Power Series;	Sec. 9.5-9.6	Notes PDF / Video / PS10
PART IV: Differential Equations			
Lec 23	Existence and Uniqueness for ODEs:	Sec. 13.11.4	Notes PDF / Video
--	Final Review	--	Review Notes PDF
--	Final Exam	--	Exam (3 hr)